Lab 7: Fuzz Testing for Input Validation Bugs

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Repository: https://github.com/AaryaTedla/L6_L7_08_ski_resort_pass

Original application output

The original application incorrectly accepts 2026/02/30 as a valid date.

```
aaryatedla@aaryatedla-IdeaPad-Pro-5-14IMH9:~/Documents/Projects/se lab/lab06$  source "/home/aaryatedla/Documents/Projects/se lab/lab06/L6_L7_08_ski_resort_pass/.venv/bin/activate"
(.venv) aaryatedla@aaryatedla-IdeaPad-Pro-5-14IMH9:~/Documents/Projects/se lab/lab06$ cd L6_L7_08_ski_resort_pass
git pull origin main
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 8 (delta 0), reused 1 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (8/8), 3.11 KiB | 795.00 KiB/s, done.
From https://github.com/AaryaTedla/L6_L7_08_ski_resort_pass
 * branch            main       -> FETCH_HEAD
   a9d4db7..8e043f2  main       -> origin/main
Updating a9d4db7..8e043f2
Fast-forward
 Lab 7/README.md         | 18 ++++++++++++++++
 Lab 7/processor.py      | 40 ++++++++++++++++++++++++++++++++++++++++
 Lab 7/test_processor.py | 17 +++++++++++++++++
 3 files changed, 75 insertions(+)
 create mode 100644 Lab 7/README.md
 create mode 100644 Lab 7/processor.py
 create mode 100644 Lab 7/test_processor.py
(.venv) aaryatedla@aaryatedla-IdeaPad-Pro-5-14IMH9:~/Documents/Projects/se lab/lab06/L6_L7_08_ski_resort_pass$ source .venv/bin/activate
cd 'Lab 7'
(.venv) aaryatedla@aaryatedla-IdeaPad-Pro-5-14IMH9:~/Documents/Projects/se lab/lab06/L6_L7_08_ski_resort_pass/Lab 7$ cp processor.py /tmp/lab7_processor_fixed.py
git show 'HEAD:Lab 7/processor.py' > processor.py
(.venv) aaryatedla@aaryatedla-IdeaPad-Pro-5-14IMH9:~/Documents/Projects/se lab/lab06/L6_L7_08_ski_resort_pass/Lab 7$ python processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2026/02/30
Pass Expiry: 2026-02-30 (VALID)

==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 250:1000
Gradient %: 25.0

==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): BEG
(.venv) aaryatedla@aaryatedla-IdeaPad-Pro-5-14IMH9:~/Documents/Projects/se lab/lab06/L6_L7_08_ski_resort_pass/Lab 7$
```

Failing fuzz tests - part 1

Original application: exceptions for malformed inputs and a failing calendar-date property.

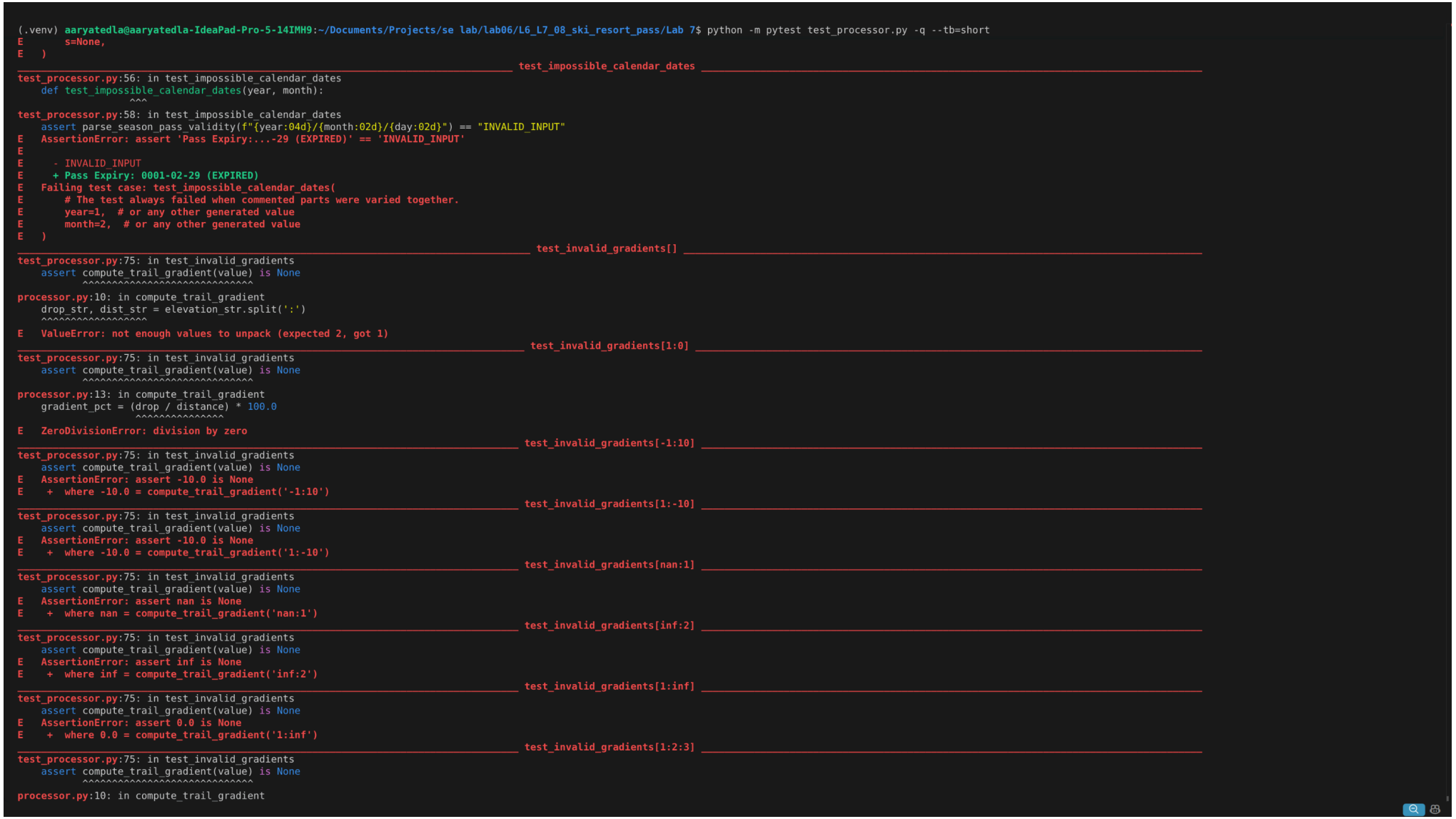
```
(.venv) aaryatedla@aaryatedla-IdeaPad-Pro-5-14IMH9:~/Documents/Projects/se lab/lab06/L6_L7_08_ski_resort_pass/Lab 7$ python -m pytest test_processor.py -q --tb=short
E   Failed: Pass validity crashed for None: AttributeError: 'NoneType' object has no attribute 'split'
E   Failing test case: test_parse_season_pass_validity_no_crash(
E       s=None,
E   )
_____ test_compute_trail_gradient_safe _____
test_processor.py:32: in test_compute_trail_gradient_safe
    result = compute_trail_gradient(s)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
processor.py:10: in compute_trail_gradient
    drop_str, dist_str = elevation_str.split(':')
                         ^^^^^^^^^^^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'split'

During handling of the above exception, another exception occurred:
test_processor.py:29: in test_compute_trail_gradient_safe
    @given(unexpected_inputs)
                   ^^^
test_processor.py:34: in test_compute_trail_gradient_safe
    pytest.fail(f"Gradient crashed for {s!r}: {type(exc).__name__}: {exc}")
E   Failed: Gradient crashed for None: AttributeError: 'NoneType' object has no attribute 'split'
E   Failing test case: test_compute_trail_gradient_safe(
E       s=None,
E   )
_____ test_calculate_lift_access_tier_safe _____
test_processor.py:42: in test_calculate_lift_access_tier_safe
    result = calculate_lift_access_tier(s)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
processor.py:17: in calculate_lift_access_tier
    prefix = zone_code.strip().upper()[:3]
             ^^^^^^^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'strip'

During handling of the above exception, another exception occurred:
test_processor.py:39: in test_calculate_lift_access_tier_safe
    @given(unexpected_inputs)
                   ^^^
test_processor.py:44: in test_calculate_lift_access_tier_safe
    pytest.fail(f"Lift access crashed for {s!r}: {type(exc).__name__}: {exc}")
E   Failed: Lift access crashed for None: AttributeError: 'NoneType' object has no attribute 'strip'
E   Failing test case: test_calculate_lift_access_tier_safe(
E       s=None,
E   )
_____ test_impossible_calendar_dates _____
test_processor.py:56: in test_impossible_calendar_dates
    def test_impossible_calendar_dates(year, month):
                   ^^^
test_processor.py:58: in test_impossible_calendar_dates
    assert parse_season_pass_validity(f"{year:04d}/{month:02d}/{day:02d}") == "INVALID_INPUT"
E   AssertionError: assert 'Pass Expiry:...-29 (EXPIRED)' == 'INVALID_INPUT'
E
E     - INVALID_INPUT
E     + Pass Expiry: 0001-02-29 (EXPIRED)
E   Failing test case: test_impossible_calendar_dates(
E       # The test always failed when commented parts were varied together.
E       year=1,  # or any other generated value
E       month=2,  # or any other generated value
E   )
```

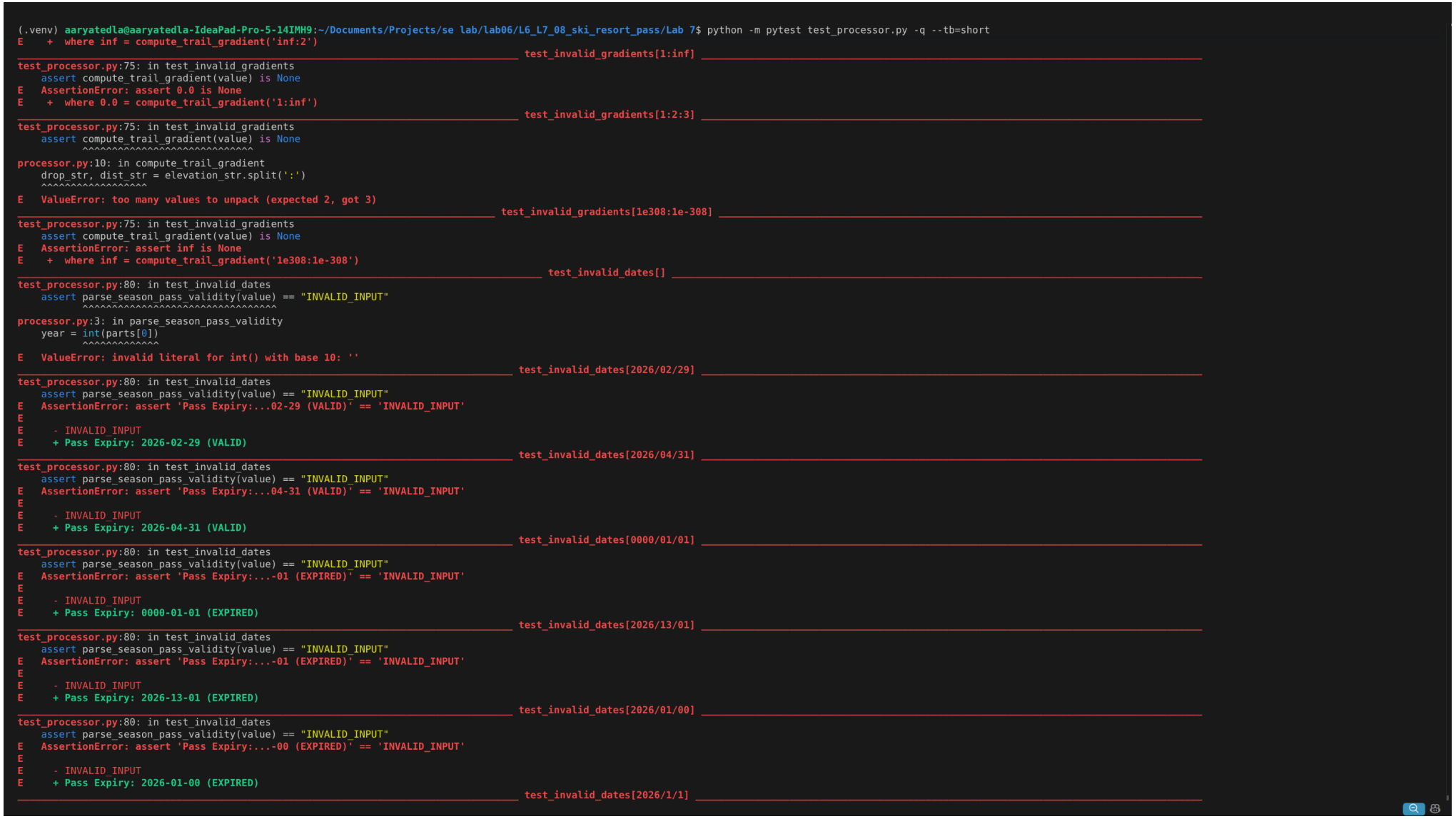
Failing fuzz tests - part 2

Invalid measurements expose zero division, negative gradients, NaN, and infinity.



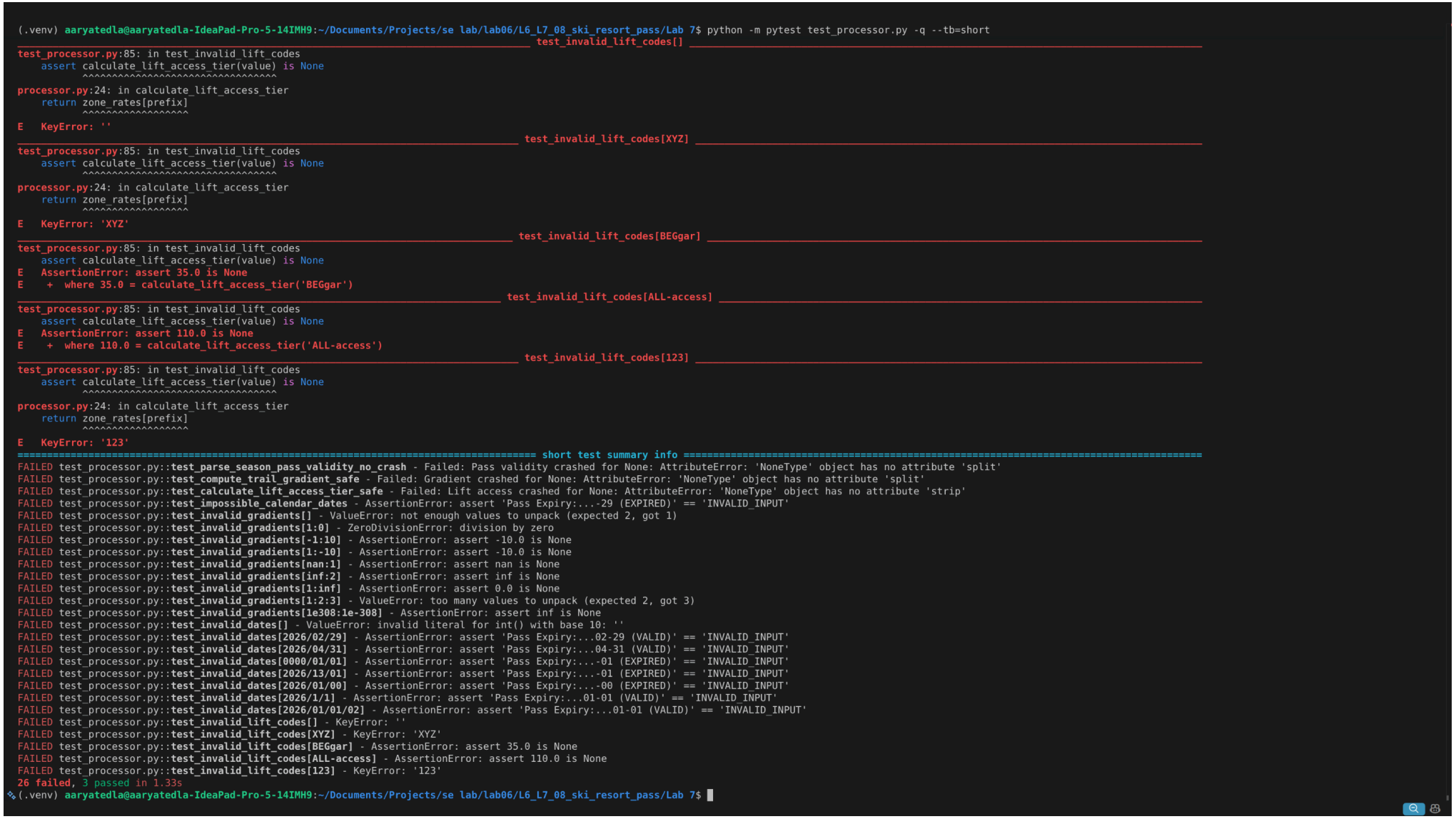
Failing fuzz tests - part 3

Further failures include overflow and invalid expiry dates.



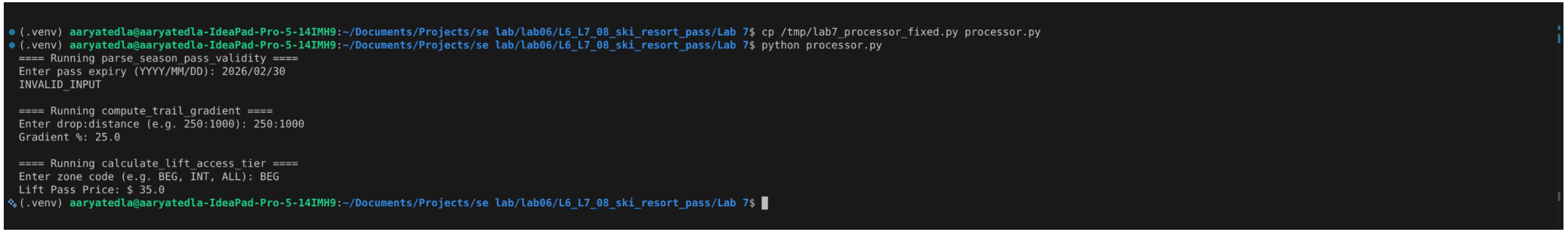
Failing fuzz tests - part 4 and summary

Before fixing: 26 failed, 3 passed.



Fixed application output

The same inputs now reject the impossible date and preserve the valid gradient and lift price.



Passing fuzz tests

After fixing: 29 passed. All screenshots are original terminal captures taken by the student.

